

CHARCOAT

PASSIVE FIRE PROTECTION

CharCoat IC



Frequently Asked Questions

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CHARCOAT IC - FREQUENTLY ASKED QUESTIONS

WHAT IS CHARCOAT IC

CharCoat IC is a RTV (Room Temperature Vulcanizing) silicone hydrophobic coating which is a specialized coating applied to surfaces to create a water-repellent (hydrophobic) layer. This coating is typically composed of silicone elastomers that cure at room temperature to form a rubber-like, flexible protective layer.

When applied to electrical insulators, RTV silicone coatings offer several benefits:

A. **Hydrophobicity:** The silicone surface repels water, preventing the formation of continuous water films on the insulator's surface. This reduces the risk of water-related conductivity issues, such as leakage currents and flashovers.

B. **Contamination Resistance:** RTV silicone coatings can maintain their hydrophobic properties even in polluted environments, where contaminants like dust, salt, or industrial pollutants would otherwise accumulate. The coating causes contaminants to bead up and be washed away by rain or wind, maintaining the insulator's performance.

C. **Improved Electrical Performance:** By minimizing the formation of conductive water films and reducing surface contamination, RTV silicone coatings help reduce leakage currents and arcing, extending the lifespan of the insulator and improving overall electrical reliability.

D. **Self-Healing:** CharCoat IC RTV silicone coating possess "Silanes" which have self-healing properties, meaning they can restore their hydrophobic characteristics even after exposure to environmental stress, making them ideal for outdoor high-voltage applications.

In summary, RTV silicone hydrophobic coatings protect electrical insulators by improving their resistance to water and pollution, enhancing their electrical performance, and reducing maintenance needs.

TECHNICAL

1. Volumetric Resistivity

The Volumetric Resistivity of CharCoat IC at 2.29×10^{15} ohm.cm well exceeds the typically required value specified of 3×10^{13} ohm.cm.

2. Surface Resistivity

The Surface Resistivity specified as " 4×10 ohm.cm" is not meaningful. CharCoat IC provides a Surface Resistivity of 1.5×10^{13} ohms.

3. Viscosity

The viscosity of CharCoat IC is around 325cP. This is the reason that it is so easy to apply and yields a smooth, even finish which is far superior to other products. The excellent surface quality achieved with CharCoat IC is a result of the formulation which utilizes a blend of solvents with differing evaporation rates.

With more viscous coatings, additional solvent often needs to be added during the course of the application process as the material becomes too thick. CharCoat IC, with its lower viscosity, completely eliminates the need to handle additional hazardous materials while on site.

4. Skin-over Time

The Skin-over and Curing times of silicone rubbers is dependent on the ambient temperature. CharCoat IC will normally become touch-dry within twenty minutes of application.

5. Dissipation Factor

Being a thin coating sprayed onto various insulating substrates this is difficult to measure and not critical to the functioning of the material. The Dielectric Strength is more important and CharCoat IC provides an exceptional breakdown voltage of 64kV/mm.

6. Tensile Strength

Once applied to an insulator surface, the coating is not subjected to any mechanical stress. For information, however, the tensile strength of CharCoat IC is approximately 6 MPa.

7. Elongation at Break

As above, it is not really relevant, but the elongation for CharCoat IC is approximately 500%

8. Tear Strength

As above, it is not really relevant, but the tear strength of CharCoat IC is approximately 25kN/m.

9. Material Density

The density of CharCoat IC is 1000kg/m³.

10. Volumetric Solids Ratio

The percentage solids by weight is a somewhat meaningless figure – it is dictated by the quantity of solvent present in the material. As the solvent evaporates, 100% solids are left on the insulator. It thus has absolutely no effect on the performance of the coating. Moreover, with lower solvent contents, “thinning/dilution is often needed during application – then the percentage solids are reduced from the “as supplied” value.

A far more important parameter is the ratio filler and other solids to the silicone content. The tendency is for manufacturers to include too much filler because it is significantly less expensive than the silicone and can improve the results of artificial tracking and erosion tests which intentionally destroy the hydrophobicity of the coating. Obviously, under service conditions, the longer hydrophobicity recovery and transfer times caused by lower silicone content, adversely affect the coating performance.

In our opinion, the ratio of silicone rubber to other solids content should be at least 55:45, and ideally 60:40.

11. Mass Solids Ratio of Material

Comments as above.

12. Coating Adhesion

Silicone rubber does not naturally adhere to other materials – hence its widespread use in the manufacture of moulds. In view of this, most coatings contain significant volumes of bonding agent. This means that the percentage of silicone in the coating is reduced and, moreover, the bulk of the bonding material is dispersed within the coating layer where it serves no beneficial purpose.

For CharCoat IC, a separate primer is provided for spraying onto the insulator prior to application of the coating. Therefore, the bonding agent exists only where it is required, i.e. at the interface between the coating and the porcelain, and no extraneous material is displacing the silicone content.

As illustrated in our bonding Test Report, after a 12 cycle, 100 hour boiling test, the specimen with a pre-applied primer achieves far better adhesion than when 10% by weight of bonding agent is added to the coating or when no primer or bonding agent is used. Further, in the “cross-hatch” test specified in ISO 16276-2, CharCoat IC achieves class 1 adhesion.

5. Insulator Cleaning

This depends entirely on the environment that the equipment finds itself in. This is why pollution severity measurement is such an important parameter. We had/have marine sites that needed cleaning or greasing every 6 months which in it self is really not practical, or industrial pollution that needed attention every year or every second year. Which ever way you look at it, they are short term solutions and coating is the way to go because you at least buy yourself 10 years maintenance free.

6. Coating vs Maintenance

The main value is definitely financial because you are comparing 10 years versus 6, 12 or 24 months maintenance. Remember that the maintenance normally requires outages and the mobilisation of many staff members and their equipment and accommodation. These costs add up over the 10 year period.

7. Creepage

Dielectric strength is measured as the maximum voltage required to produce a dielectric breakdown through a material. So if the creepage goes up so does the dielectric strength, and vice versa. This must not be confused with the breakdown of air between two components.

8. Application

Please refer to our CharCoat IC Application guide for full information.

CharCoat IC is designed to be applied over suitably prepared and cleaned insulators.

A thorough cleaning of the insulator is required. If the surface is just dusty or if close to the ocean with plain salt deposits, normal clean water will suffice for cleaning. If, however, the insulators are seriously contaminated we recommend Toluene or Xylene for cleaning down.

Spray Guns and Tips

Recommend 208 airless tip for the CharCoat IC Primer at 60 bar and a more than satisfactory finish with a 412 tip at 100 bar for the CharCoat IC coating using a Contractor Inline gun or similar. Alternatively, application can be completed on site with compressors and High-Pressure hoses and spray guns.

